

# RAHUL MITRA

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## EDUCATION

### National University of Singapore

Aug 2023 – Jul 2027

#### Bachelor of Engineering, Industrial Systems Engineering

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.

## WORK EXPERIENCE

### Operations Research (NUS System Design Project), Contract, STMicroelectronics

Aug 2026 – Present

- Design an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, replacing experience-based slotting with capacity-aware decisions drawn from 3 live inputs: picking history, demand forecasts, and available capacity.

### Robotics Vision Engineer, Amazon (Blended AI+X)

Jul 2026 – Present

- Build vision systems for 3 imaging workstreams: camera ISP enhancement, super-resolution, and image restoration under motion and low light.
- Train UNet and other CNN architectures for single-image super-resolution, upscaling frames 4 times and lifting PSNR and SSIM so a low-cost camera reaches frame quality on par with premium sensors.

### Operational AI Systems & Data Engineer (Contract), Abbott Laboratories

Jun 2026 – Present

- Assemble a 5-stage offline process-intelligence platform for globally distributed spray-drying plants: GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain over a locally hosted Gemma model.

### Platform & Solutions Engineer (Sparks Citizen Developer), People's Association

Sep 2025 – Sep 2026

- Delivered a digital garden-plot balloting platform across admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.

### Data Analytics & Product Management Intern, Abbott Laboratories

Jan 2026 – Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, then compared 3 approaches (heuristic, MIP, and genetic algorithm) to turn production objectives and constraints into decision-ready schedules.
- Trained regional engineering teams across Asia and Europe on repeatable AI workflows spanning 7 areas: prompting, model selection, benchmarks, cost awareness, agents, validation, and safeguards.

## PROJECTS

### Team Lead & Finalist, Swarmline: SG Defense Tech Hackathon (1,300+ applicants)

Sep 2026

- Directed a 5-person team to architect decentralised drone-swarm coordination on correlated random-walk search, gradient-flow dispersion, delay-tolerant gossip, cross-inhibition consensus, and response-threshold task allocation.

### Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams)

Jan 2026

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once, placing top 8 of 50+ teams.

### Team Lead, Maritime Deficiency Severity Forecasting: Maritime Hackathon

Jun 2025

- Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers and gradient clipping on the ASPIRE 2A supercomputer.

## LEADERSHIP AND ACTIVITIES

### NTUC Health Active Ageing Centre, Elderly Care Volunteer

Sep 2025 – Sep 2026

- Coordinate weekly senior-engagement activities, building empathy for healthcare, accessibility, and community-focused product design.

## SKILLS AND CERTIFICATIONS

Technical: Python, PyTorch, BERT, DNN, RAG, AI agents, GPT-4, Gemini, Claude Code, ChatGPT, Codex, ComfyUI, SEALION, ONNX, Chroma, local LLM serving (llama.cpp), vector search, model evaluation, NLP; vision: camera ISP, super-resolution, image restoration, camera models and calibration, fundamental and essential matrices, epipolar geometry, absolute pose estimation, three-view geometry, structure-from-motion (SfM) with bundle adjustment, two- and multi-view stereo, generalised camera models; HPC: ASPIRE 2A, CUDA; deep RL (PPO, A2C, DDPG, DQN)  
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery; Reinforcement Learning and Deep RL Python (Packt)